

Mollifiers

following Evan's *Partial Differential Equations* Appendix C.4

Mollifiers allow us to create C^∞ functions.

DEF 1. Define the function,

$$\eta: \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto \begin{cases} C \exp\left(\frac{1}{|\mathbf{x}|-1}\right), & |\mathbf{x}| < 1 \\ 0, & |\mathbf{x}| \geq 1 \end{cases}$$

Where C is s.t. $\int_{\mathbb{R}^n} \eta \, dV = 1$. η is known as the ‘standard mollifier’ and is $C^\infty(\mathbb{R}^n)$.

DEF 2. We may dilate η by defining,

$$\eta_\varepsilon(\mathbf{x}) := \frac{1}{\varepsilon^n} \eta\left(\frac{\mathbf{x}}{\varepsilon}\right)$$

So the support of η_ε is $\{|\mathbf{x}| \leq \varepsilon\}$. η_ε are known as ‘mollifiers’ and the functions η_ε are $C^\infty(\mathbb{R}^n)$.

DEF 3. For a function $f: U \rightarrow \mathbb{R}^n$ for $U \subseteq \mathbb{R}^n$ an open set define its ‘mollification’,

$$\eta_\varepsilon f := \eta_\varepsilon * f \Leftrightarrow \eta_\varepsilon f(\mathbf{x}) = \int_U \eta_\varepsilon(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) \, dV_{\mathbf{y}} = \int_{\mathbb{B}_\varepsilon^n} \eta_\varepsilon(\mathbf{y}) f(\mathbf{x} - \mathbf{y}) \, dV_{\mathbf{y}}$$

Whereby $\eta_\varepsilon f: U_\varepsilon \rightarrow \mathbb{R}$, where $U_\varepsilon = \{\mathbf{x} \in U \mid d(\mathbf{x}, \partial U) < \varepsilon\}$.

THM 1. PROPERTIES OF MOLLIFIERS

We have the following results:

1. $\eta_\varepsilon f \in C^\infty(U_\varepsilon)$
2. $\eta_\varepsilon f \rightarrow f$ a.e. as $\varepsilon \rightarrow 0$
3. If $f \in C(U)$ then $\eta_\varepsilon \rightrightarrows \eta$ on compact subsets of U
4. For $p \in [1, \infty)$ and $f \in L^p_{\text{loc}}(U)$, $\|\eta_\varepsilon f - f\|_p \rightarrow 0$

Proof.

(i)

Let $\mathbf{x} \in U_\varepsilon$ and h sufficiently small so that $\mathbf{x} + h\mathbf{e}_i \in U_\varepsilon$. We have,

$$\frac{\eta_\varepsilon f(\mathbf{x} + h\mathbf{e}_i) - \eta_\varepsilon f(\mathbf{x})}{h} = \frac{1}{\varepsilon^n} \int_U \frac{1}{h} \left(\eta\left(\frac{\mathbf{x} + h\mathbf{e}_i - \mathbf{y}}{\varepsilon}\right) - \eta\left(\frac{\mathbf{x} - \mathbf{y}}{\varepsilon}\right) \right) f(\mathbf{y}) \, dV_{\mathbf{y}}$$

The value of the integral is equal to the value of the integral over the interior of U . In this interior we have,

$$\frac{1}{h} \left(\eta\left(\frac{\mathbf{x} + h\mathbf{e}_i - \mathbf{y}}{\varepsilon}\right) - \eta\left(\frac{\mathbf{x} - \mathbf{y}}{\varepsilon}\right) \right) \Rightarrow \frac{1}{\varepsilon} \frac{\partial \eta}{\partial x_i} \left(\frac{\mathbf{x} - \mathbf{y}}{\varepsilon} \right)$$

At which,

$$\frac{1}{\varepsilon^n} \int_U \frac{1}{h} \left(\eta \left(\frac{\mathbf{x} + h\mathbf{e}_i - \mathbf{y}}{\varepsilon} \right) - \eta \left(\frac{\mathbf{x} - \mathbf{y}}{\varepsilon} \right) \right) f(\mathbf{y}) \, dV_{\mathbf{y}} \rightarrow \int_U \frac{\partial \eta_\varepsilon}{\partial x_i}(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) \, dV_{\mathbf{y}}$$

So $\partial_i \eta_\varepsilon f$ exists. This argument is the same for any derivative ∂_α in which,

$$\frac{\partial^\alpha}{\partial x_\alpha} \eta_\varepsilon f = \int_U \frac{\partial^\alpha \eta_\varepsilon}{\partial x_\alpha}(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) \, dV_{\mathbf{y}} \quad (2)$$

Recall Lebesgue's Differentiation Theorem,

$$\frac{1}{|B_r^\mathbf{x}|} \int_{B_r^\mathbf{x}} |f(\mathbf{y}) - f(\mathbf{x})| \, dV_{\mathbf{y}} \rightarrow 0$$

Hence, as mollifiers integrate to unit,

$$\begin{aligned} |\eta_\varepsilon f(\mathbf{x}) - f(\mathbf{x})| &= \left| \int_{B_\varepsilon^\mathbf{x}} \eta_\varepsilon(\mathbf{x} - \mathbf{y}) (f(\mathbf{y}) - f(\mathbf{x})) \, dV_{\mathbf{y}} \right| \\ &\leq \frac{1}{\varepsilon^n} \int_{B_\varepsilon^\mathbf{x}} \eta \left(\frac{\mathbf{x} - \mathbf{y}}{\varepsilon} \right) |f(\mathbf{y}) - f(\mathbf{x})| \, dV_{\mathbf{y}} \\ &\leq \frac{1}{|B_\varepsilon^\mathbf{x}|} \int_{B_\varepsilon^\mathbf{x}} |f(\mathbf{y}) - f(\mathbf{x})| \, dV_{\mathbf{y}} \quad (\eta \leq 1) \\ &\rightarrow 0 \end{aligned} \quad (3)$$

Let $f \in C(U)$. Given a compact subset $V \subseteq U$ the a.e. pointwise convergence of (2) is uniform in the interior.

(4)

Let $p \in [1, \infty)$ and $f \in L_{\text{loc}}^p(U)$. Let V be a compact subset and in its interior we claim $\|\eta_\varepsilon f\|_p \leq \|f\|_p$. We first show this in the case $p \neq 1$,

$$\begin{aligned} |\eta_\varepsilon f(\mathbf{x})| &= \left| \int_{B_\varepsilon^\mathbf{x}} \eta_\varepsilon(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) \, dV_{\mathbf{y}} \right| \\ &\leq \int_{B_\varepsilon^\mathbf{x}} |\eta^{1-1/p}(\mathbf{x} - \mathbf{y})| |\eta^{1/p}(\mathbf{x} - \mathbf{y}) f(\mathbf{y})| \, dV_{\mathbf{y}} \\ &\leq \left(\int_{B_\varepsilon^\mathbf{x}} \eta_\varepsilon(\mathbf{x} - \mathbf{y}) \, dV_{\mathbf{y}} \right)^{1-1/p} \left(\int_{B_\varepsilon^\mathbf{x}} \eta_\varepsilon(\mathbf{x} - \mathbf{y}) |f(\mathbf{y})|^p \, dV_{\mathbf{y}} \right)^{1/p} \quad (\text{Holder's Inequality}) \\ &= \left(\int_{B_\varepsilon^\mathbf{x}} \eta_\varepsilon(\mathbf{x} - \mathbf{y}) |f(\mathbf{y})|^p \, dV_{\mathbf{y}} \right)^{1/p} \quad \left(\int_{\mathbb{B}_\varepsilon^n} \eta = 1 \right) \end{aligned}$$

Hence,

$$\int_V |\eta_\varepsilon f(\mathbf{x})|^p dV_{\mathbf{x}} \leq \int_V \int_{B_\varepsilon^{\mathbf{x}}} \eta_\varepsilon(\mathbf{x}-\mathbf{y}) |f(\mathbf{y})|^p dV_{\mathbf{y}} dV_{\mathbf{x}} = \int_V |f(\mathbf{y})|^p \int_{B_\varepsilon^{\mathbf{y}}} \eta_\varepsilon(\mathbf{x}-\mathbf{y}) dV_{\mathbf{x}} dV_{\mathbf{y}} = \int_V |f(\mathbf{y})|^p dV_{\mathbf{y}}$$

Therefore fix $\delta > 0$ and choose $g \in C(U)$ s.t. $\|f - g\|_p < \delta$. By Minkowski's Inequality,

$$\begin{aligned} \|\eta_\varepsilon f - f\|_p &\leq \|\eta_\varepsilon f - \eta_\varepsilon g\|_p + \|\eta_\varepsilon g - g\|_p + \|g - f\|_p \\ &\leq 2\|f - g\|_p + \|\eta_\varepsilon g - g\|_p \\ &\leq 2\delta + \|\eta_\varepsilon g - g\|_p \\ &\rightarrow 2\delta \end{aligned}$$

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